The Voxel T-Matrix

Discretising an elastic scatterer into cubes: the matched propagator, and the accuracy of the result against exact Mie

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Abstract

A scatterer of arbitrary shape can be represented as a lattice of small cubes, each carrying an analytic single-site T-matrix, coupled through the elastodynamic propagator and solved by Foldy–Lax. This note sets out that construction, identifies the condition the propagator must satisfy for the scheme to be consistent, and measures the accuracy that results.

The condition is a pairing. A single-site T-matrix closes on a particular functional of the internal field — the value at the cube centre, or the average over the cube — and the propagator it is used with must close on the same one. The collocation closure requires the propagator averaged over the *receiver cell only*: one power of the cell form factor, not the two of a Galerkin pairing and not the none of a point evaluation.

That average is given in closed form. It is regular at every non-zero lattice separation, its far field is a scalar multiple of the point propagator per mode, and near contact it needs a quadrature order that follows the separation.

Accuracy is measured against the exact elastic Mie solution for a sphere, which is the one arbiter that is both exact and laterally varying. Two controls are needed for the comparison to measure the method rather than the test: the observation distance must reach the far field, and the staircase volume of the voxelised body must be divided out. With those in place the matched pairing is 4.7–8.0× closer to Mie than a point propagator in the Rayleigh regime. At $ka = 0.5$ the staircase of the sharp sphere decides the error and the two propagators cannot be told apart there; on a smoothly graded sphere, which has no staircase, the matched pairing converges at second order to the exact solution at $k_S a = 0.5$ and 1, while the point propagator stalls near 0.9% however fine the grid.

These numbers supersede two earlier sets: one measured against an arbiter that carried three compensating sign errors, and one whose far field counted the incident wave's linear variation twice. §5 records what changed.

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1 The construction

1.1 Foldy–Lax on a lattice of cubes

Partition the scatterer into N cubes of side d on a regular lattice, with centres $\mathbf{r}_m$. Each cube is represented by a local T-matrix $\mathbf{T}_{\mathrm{loc}}$ acting on a nine-component state — three displacements and six Voigt strains,

$$\boldsymbol{\psi} = \left(u_z,\, u_x,\, u_y,\, \varepsilon_{zz},\, \varepsilon_{xx},\, \varepsilon_{yy},\, 2\varepsilon_{xy},\, 2\varepsilon_{zy},\, 2\varepsilon_{zx}\right)^{\mathsf{T}}, \tag{1}$$

which is the smallest state closed under the coupling: the source a cube radiates depends on both the displacement and the strain it experiences.

The exciting field at cube m is the incident field plus the field radiated by every other cube,

$$\boldsymbol{\psi}_m = \boldsymbol{\psi}_m^{\mathrm{inc}} + \sum_{n \neq m} \mathbf{P}(\mathbf{r}_m - \mathbf{r}_n)\, \mathbf{T}_{\mathrm{loc}}\, \boldsymbol{\psi}_n, \qquad \text{that is} \qquad (\mathbf{I} - \mathbf{P}\mathbf{T})\boldsymbol{\psi} = \boldsymbol{\psi}^{\mathrm{inc}}, \tag{2}$$

with $\mathbf{P}(\mathbf{s})$ the 9×9 propagator between cubes at separation $\mathbf{s}$. The composite T-matrix of the whole body follows by summing the sources, $\mathbf{T}_{\mathrm{comp}} = \sum_m \mathbf{T}_{\mathrm{loc}} \boldsymbol{\psi}_m$.

The self term is excluded from the sum: a cube's action on itself is contained in $\mathbf{T}_{\mathrm{loc}}$, which is where the singular part of the Green's tensor and its static (Eshelby) content live.

1.2 The propagator

P is assembled from the elastodynamic Green's tensor and its derivatives, in blocks

$$\mathbf{P} = \begin{pmatrix} \mathbf{G} & \mathbf{C} \\ \mathbf{H} & \mathbf{S} \end{pmatrix}, \tag{3}$$

where **G** maps force to displacement, **S** maps stress source to strain, and **C**, **H** are the cross couplings obtained by Voigt contraction of ∂G and $\partial\partial G$. Written through the Kupradze representation, every block is built from the two scalar Helmholtz kernels g_P and g_S and their Cartesian derivative tensors, so the P and S content is separable — which §3 requires.

2 The pairing condition

2.1 Two closures

Volume-averaging the Lippmann–Schwinger equation with a uniform internal ansatz gives

$$\langle \nabla \mathbf{u} \rangle = \langle \nabla \mathbf{u}^0 \rangle + \left[\frac{1}{V} \iint \partial\partial G(\mathbf{x} - \mathbf{x}')\, d\mathbf{x}\, d\mathbf{x}' \right] \Delta c\, E, \tag{4}$$

a *double* average, while evaluating the same equation at the cube centre gives

$$\nabla \mathbf{u}(0) = \nabla \mathbf{u}^0(0) + \left[\int_V \partial\partial G(-\mathbf{x}')\, d\mathbf{x}' \right] \Delta c\, E, \tag{5}$$

a *single* average. Neither is wrong: one closes on the volume average of the internal field, the other on its centre value. For an ellipsoid they coincide, because Eshelby's uniformity theorem makes the internal field uniform; for a cube they differ.

2.2 The condition

A T-matrix built on (5) must be used with a propagator that also closes on the centre value. The propagator between two cubes therefore carries *one* power of the cell form factor — the average over the receiver cell, with the source treated as a point — and not two. Writing the cell indicator's transform,

$$\text{single average} \quad \longleftrightarrow \quad \operatorname{sinc}(k_x h) \operatorname{sinc}(k_y h) \operatorname{sinc}(k_z h), \qquad h = d/2, \tag{6}$$

against the squared factor that a Galerkin pairing produces. A point propagator carries neither and is not a consistent partner for either closure.

Per mode. The form factor must be applied to each mode separately. The P and S poles carry different k_z at the same lateral wavenumber, so a single scalar factor applied to the assembled 9×9 is incorrect.

3 The single cell average

3.1 Definition and regularity

For cubes at separation $\mathbf{s}$ on the lattice, the matched propagator is

$$\langle \mathbf{P} \rangle(\mathbf{s}) = \frac{1}{d^3} \int_{|\mathbf{u}|_\infty \le h} \mathbf{P}(\mathbf{s} - \mathbf{u})\, d\mathbf{u}, \qquad h = d/2. \tag{7}$$

With a point source and a cubic receiver cell, the closest approach of the cell to the source is its near face, so

$$|\mathbf{s} - \mathbf{u}| \;\ge\; d/2 \qquad \text{for every lattice separation } \mathbf{s} \neq 0, \tag{8}$$

and the integrand in (7) is regular everywhere on the domain. The double average has no such property: its integrand reaches $r = 0$ whenever the cells touch.

3.2 The far field, in closed form

The midpoint error of a cell average is

$$\langle g \rangle - g = \frac{d^2}{24} \nabla^2 g + O(d^4), \tag{9}$$

and for a Helmholtz kernel $\nabla^2 g_c = -\kappa_c^2 g_c$ exactly away from the origin. Derivatives commute with ∇^2, so the same holds for every derivative tensor and hence for every block of (3). Therefore, away from contact,

$$\langle \mathbf{P} \rangle = \Big(1 - \frac{\kappa_c^2 d^2}{24}\Big)\mathbf{P} + O(d^4) \qquad \text{per mode } c \in \{P, S\}, \tag{10}$$

the two modes carrying their own κ and therefore not combinable before the average is taken.

3.3 Quadrature near contact

Inside the near shell (10) is not accurate enough and (7) is integrated directly. The integrand is regular but peaked near the face of the cell closest to the source, and a Gauss rule converges slowly against a nearby peak. The difficulty is a property of the separation alone,

$$q(\mathbf{s}) = \frac{\big|\max(|\mathbf{s}| - h,\, 0)\big|}{d}, \tag{11}$$

componentwise in the clamp, which takes the values 0.5 for a face neighbour, $1/\sqrt{2}$ for an edge neighbour and $\sqrt{3}/2$ for a corner one. Convergence of a product Gauss rule against a 24-point reference:

neighbour	$g = 4$	$g = 8$	$g = 12$	$g = 16$
face ($q = 0.50$)	1.8×10^{-2}	2.9×10^{-5}	3.6×10^{-8}	1.4×10^{-10}
edge ($q = 0.71$)	6.5×10^{-4}	9.0×10^{-8}	1.6×10^{-10}	—
corner ($q = 0.87$)	6.1×10^{-5}	2.3×10^{-10}	—	—

Table 1. The order should follow q: 16 points per axis for face neighbours, 12 for edge, 8 for corner, and the base order elsewhere. Only six and twelve separations respectively are expensive, so the adaptive rule costs almost nothing.

Two independent evaluations of (7) — direct quadrature and the closed form (10) — agree to $O(d^4)$, halving d reducing their difference by sixteen (5.3×10^{-4}, 3.4×10^{-5}, 2.1×10^{-6} at $d = 0.2,\ 0.1,\ 0.05$), and the average reduces to the point propagator at second order as (9) requires.

3.4 Cost

P depends only on the separation, so on a regular lattice there are $O(n^3)$ distinct blocks rather than $O(N^2)$ pairs. Building the kernel once per distinct separation — which is what an FFT-accelerated solver does anyway — makes the averaged propagator affordable: the few near-contact separations needing high-order quadrature are evaluated once, whatever the cell count.

4 Measuring the accuracy

The exact elastic Mie solution for a homogeneous sphere is the natural arbiter: it is exact, and unlike a layered reference it is laterally varying. Two things must be controlled for the comparison to measure the discretisation rather than the test.

4.1 The observation distance

If the scattered field is compared through a far-field asymptotic on one side and an exact expression on the other, the neglected $1/(kr)$ term must be small compared with the quantity being measured. It is not a weak condition. Measured at $ka = 0.1$ with the sphere divided 4 per edge:

r/a	$k_S r$	volume-corrected error
100	10	0.5043
500	50	0.0575
5×10^3	500	0.0075
5×10^4	5000	0.0068

Table 2. The comparison is usable only from $k_S r \sim 500$; at $k_S r = 50$ it overstates the error sixfold. It is not fully converged even at the top: the last decade still moves the answer by 10%, which sets the precision of every small number in §5. The earlier version of this table read 0.0088 at both of the last two rows and so looked converged; correcting the arbiter made the errors smaller without making the harness better, and the residual $1/(kr)$ term is now a visible fraction of them.

4.2 The staircase

A cubic lattice does not fill a sphere, and its coverage is not monotonic in the number of cells per edge. For $n_{\mathrm{sub}} = 3, 4, 6$ the voxel volume exceeds the sphere's by 26%, 4.7% and 17%, and the raw error tracks that dilution to within a factor 1.4. Two consequences follow.

First, the raw error must be divided by the volume ratio before it says anything about the propagator. Second, a refinement ladder in n_{sub} measures the geometry, not the method: the error moves non-monotonically because the dilution does. The comparison that isolates the discretisation is at *fixed* geometry — the same voxel set, the same staircase, one propagator exchanged for the other — and its stability across resolutions is what makes it meaningful.

5 Results

> *What changed, and why. The numbers of the first version compared two objects that were* both *wrong. The exact Mie solution carried a spurious* $(-1)^n$ *in its multipole sum, inverting the density/dipole channel; the Foldy–Lax far field carried a sign error in its force term; and the contrast-extraction route carried a third error that cancelled the first. They were found by an impedance-null test and fixed together on 20 September 2026. On 29 September a second far-field defect was fixed: it counted the incident wave's linear variation twice, an error of order* $(ka)^2$ *that no refinement removes. The* $ka = 0.5$ *"reversal" reported after the first fix (the averaged propagator* 0.71–$0.84\times$ *as good as the point one) did not survive it, and the* $ka = 1.5$ *numbers of that version are withdrawn. On 31 October 2026 the single site became the uniform-strain closure exact at every frequency, in place of a cube T-matrix with far-field form-factor corrections; on this sharp sphere that moves the errors by under a tenth at* $ka = 0.1$ *and by up to a half at* $ka = 0.5$*, without changing either verdict. What survives throughout is §2–§3, which concern the propagator alone and never passed through the arbiter.*

Volume-corrected far-field error against exact Mie, for a sphere with $\Delta\lambda = 2$, $\Delta\mu = 1$, $\Delta\rho = 0.1$ on a background $\alpha = 5$, $\beta = 3$, $\rho = 2.5$ (`scripts/gate_sphere_cell_average_vs_mie.py`):

ka	n_{sub}	point propagator	matched average	ratio
0.1	3	0.0056	**0.0012**	4.66
	4	0.0072	**0.0010**	7.36
	6	0.0087	**0.0011**	8.01
0.5	3	**0.0055**	0.0092	0.60
	4	0.0098	**0.0028**	3.48
	6	0.0084	**0.0062**	1.35

Table 3. Bold is the better of the pair. In the Rayleigh regime the matched average is 4.7 to 8.0 times closer to Mie, by a factor that varies by 50% across resolutions. At $ka = 0.5$ the ratios straddle one: on a staircased sphere no verdict is available there.

5.1 The pairing on a body without a staircase

The smoothly graded sphere of radius $a = 10\,\mathrm{m}$, its contrast full in the core $r < a/2$ and falling to zero across the shell as a C^2 smoothstep, has no staircase: the contrast and its slope vanish at the surface, so the voxelised body converges to the body. Solved with each cell at the profile's centre value and the exact closure as its single site, against its exact solution, the far field's error is (`scripts/measure_pairing_graded_sphere.py`):

	$k_S a = 0.5$		$k_S a = 1$	
n_{sub}	point	matched average	point	matched average
6	7.5×10^{-3}	2.8×10^{-3}	7.6×10^{-3}	1.6×10^{-3}
8	8.1×10^{-3}	1.6×10^{-3}	8.5×10^{-3}	1.3×10^{-3}
12	8.8×10^{-3}	1.8×10^{-4}	9.4×10^{-3}	1.6×10^{-4}
16	9.1×10^{-3}	3.5×10^{-5}	9.7×10^{-3}	3.5×10^{-5}
24	9.2×10^{-3}	1.3×10^{-5}	9.9×10^{-3}	1.2×10^{-5}

Table 4. With the point propagator the scheme does not converge: its error stays near 0.9% and grows slightly as the grid is refined. With the average over the receiving cell it converges at second order to the exact solution, 700 to 800 times closer at 24 cells across.

A plausible reading, not derived here: the two propagators differ at separation s by $(d^2/24)\nabla^2 G$ to leading order, and summed over the neighbours of a cell the δ-function content of $\nabla^2 G$ contributes an amount that does not shrink with d. Whatever the mechanism, the measurement shows that the pairing of §2 is not a Rayleigh-regime refinement but the condition for the scheme to converge to the continuum at all.

5.2 The three regimes

Rayleigh, $ka \lesssim 0.1$. On the sharp sphere the matched average reaches 0.10–0.12%, at the shape-factor floor of representing a sphere by cubes, against 0.56–0.87% for the point propagator.

Transition, $ka \sim 0.5$ to 1. On the sharp sphere the staircase decides the error and the two propagators cannot be told apart; on the graded sphere the matched pairing converges and the point propagator does not (above). The question the first version left open, whether the collocation closure loses validity as ka_{sub} grows, is answered: at these frequencies it does not.

Resonance, $ka \gtrsim 1$. On a sphere with a sharp surface the error is set by the representation of the curved boundary at a wavelength comparable with the radius, not by the propagator. The $ka = 1.5$ numbers of the first version, 22–28%, were measured with the far field that counted the incident wave twice and are not repeated here.

6 Summary

1. Partition the body into cubes; give each the single-site T-matrix of the uniform-strain closure, its self term exact at every frequency.
2. Couple them with the propagator averaged over the *receiver* cell, (7), applied per mode — matching the collocation closure of the single-site T-matrix.
3. Use the closed form (10) beyond a near shell and direct quadrature inside it, with the order chosen from (11).
4. Solve (2); build the kernel once per distinct separation.

5. When comparing against an exact reference, reach the far field and divide out the staircase volume; compare at fixed geometry.

On a sharp sphere the method is accurate to about a tenth of a per cent in the Rayleigh regime and is limited by the staircase of the curved surface beyond it, where a representation that follows the boundary is required instead.

The pairing of step 2 is the condition for the scheme to converge at all: on a body without a staircase the matched pairing converges at second order to the exact solution at $k_S a = 0.5$ and 1, while the point propagator stalls near 0.9%.